

L13 — Metrics & Applications

Evaluation, ethics, and case studies

Study summary generated from lecture slides. ETH AI for Digital Characters.

TTS evaluation (recap)

- MOS, A/B tests, DRT, MRT, WER for intelligibility

Responsible AI

- Harmful system: negative real-world impact
- Biased system: different performance across subpopulations
- Bias sources: training data, system design, distribution shift
- ASR racial disparities; face recognition demographic bias
- Problem formulation should be ethical (e.g. predict criminality from face = bad task)

Case studies

- Digital Einstein: image generation, upcoming leg/feet animation
- Virtual Doctor: experimental user study setup
- Virtual Psychotherapist: human vs AI therapist evaluation

Exam preparation

- Review all pipeline components and evaluation metrics
- Know trade-offs: rule-based vs LLM, concatenative vs neural TTS, etc.